FUZE BIOTECH

FTP F1 (FUZE Textile Product)

Technical Data Sheet

Product Description

FTP F1 (FUZE Textile Product) is a textile treatment designed for integration during standard textile finishing operations. It can be applied using conventional pad-bath, exhaust, or spray processing equipment. The treatment integrates within the textile structure while maintaining the original fabric characteristics. FTP F1 is the highest-performance tier in the FUZE product range, targeting 1.0 mg/kg active loading on treated fabric for antimicrobial performance.

Product ID: FTP F1 **Document ID:** TDS-FTP-F1-002 **Revision:** 1.0

Product Properties

Property	Value
Appearance	Clear, colorless liquid
Carrier	18 megaohm DI water (99.997% w/w)
Active concentration	30 mg/L (0.003% w/w)
pH	~7 (neutral)
Odor / Flavor	None / None
Reactivity	Non-reactive with textile finishing chemistries
Solubility	Fully miscible in water
Viscosity	Similar to water (~1.0 cP)
Density	~1.0 g/mL
Packaging	19 L HDPE container

Recommended Product Concentration in Textile Recipe

Pad Application — 80% Dry-to-Wet Pickup (typical pad process)

Tier	Target (mg/kg WOF)	Bath Conc. (mg/L)	Dosage (mL/L bath)	per 100 L Bath
F1	1.00	1.25	41.7	4.17 L
F2	0.75	0.94	31.2	3.12 L
F3	0.50	0.62	20.8	2.08 L
F4	0.25	0.31	10.4	1.04 L

Pad Application — 15% Wet-to-Wet Pickup (pre-wetted fabric)

Tier	Target (mg/kg WOF)	Bath Conc. (mg/L)	Dosage (mL/L bath)	per 100 L Bath
F1	1.00	6.67	222.2	22.22 L
F2	0.75	5.00	166.7	16.67 L
F3	0.50	3.33	111.1	11.11 L
F4	0.25	1.67	55.6	5.56 L

Calculation method: Bath concentration (mg/L) = Target loading (mg/kg) ÷ (pickup % ÷ 100). Dosage (mL per liter of bath) = Bath concentration (mg/L) ÷ stock concentration × 1000. Dosage per 100 L bath = Dosage (mL/L) × 100 ÷ 1000. Reference fabric 100 GSM; GSM cancels out of the bath-concentration formula. Adjust for fabric weight/construction and confirm by mill trial.

Fabric Compatibility

FTP F1 can be applied to polyester, nylon, polyester blends, cotton blends, synthetic performance fabrics, and down-proof fabrics. Compatibility with specialty finishing chemistries should be confirmed through standard mill trials.

Application Methods

Pad-Dry (Continuous Finishing)

Prepare bath with water carrier and add FTP F1 at the dosage indicated for the target loading and expected pickup. Pad fabric to achieve 10–90% wet pickup depending on fabric construction. Dry fabric using standard mill drying conditions. No curing step required.

Exhaust (Batch Processing)

Add FTP F1 to the dyebath or a dedicated finishing bath at the appropriate concentration. Run at standard dyeing temperature and time. FUZE integrates during the exhaust cycle. Compatible with existing dyebath chemistry — no process modification required.

Spray Application

Apply using spray heads spaced at 6-inch intervals across fabric width (top and bottom). Standard line speed: 15 m/min. Adjust spray concentration based on pump rate and target dosage. Suitable for continuous finishing lines.

Storage and Handling

Storage: Store in original HDPE container, tightly closed. Temperature: 5°C–35°C. Avoid freezing.
Shelf life: 12 months when stored unopened.
Handling: Use standard industrial hygiene practices. No special PPE required under normal conditions.

Regulatory Information

FTP F1 contains elemental silver (CAS 7440-22-4) at 0.003% w/w. Not classified as hazardous under GHS at supplied concentration. Consult the Safety Data Sheet (SDS) for full regulatory and safety information.

FTP F1 is manufactured, delivered, applied, and stored exclusively as a dilute aqueous dispersion. The active is never supplied or made available in dry or free particulate form, and is not present on the finished textile as free nano-scale material available for interaction or release. On this basis the product is positioned as a non-nano textile treatment for regulatory purposes.

Contact

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